

WASTE CLASSIFICATION USING IMAGE CLUSTERING

A PROJECT REPORT

Course: MACHINE LEARNING PRACTICAL

Project Title: Waste Classification Using Image Clustering

Laboratory number: LAB 14, Complex Computing Activity

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1 INTRODUCTION:

In urban areas, especially in Pakistan. Waste accumulation is inevitable and it is a prominent issue as waste can be seen gathered in alleys and corners. Thus, sorting of such waste into Recyclable, biodegradable and non-recyclable may aid in dealing with this problem. Making our environment cleaner and might even reduce the creation of waste.

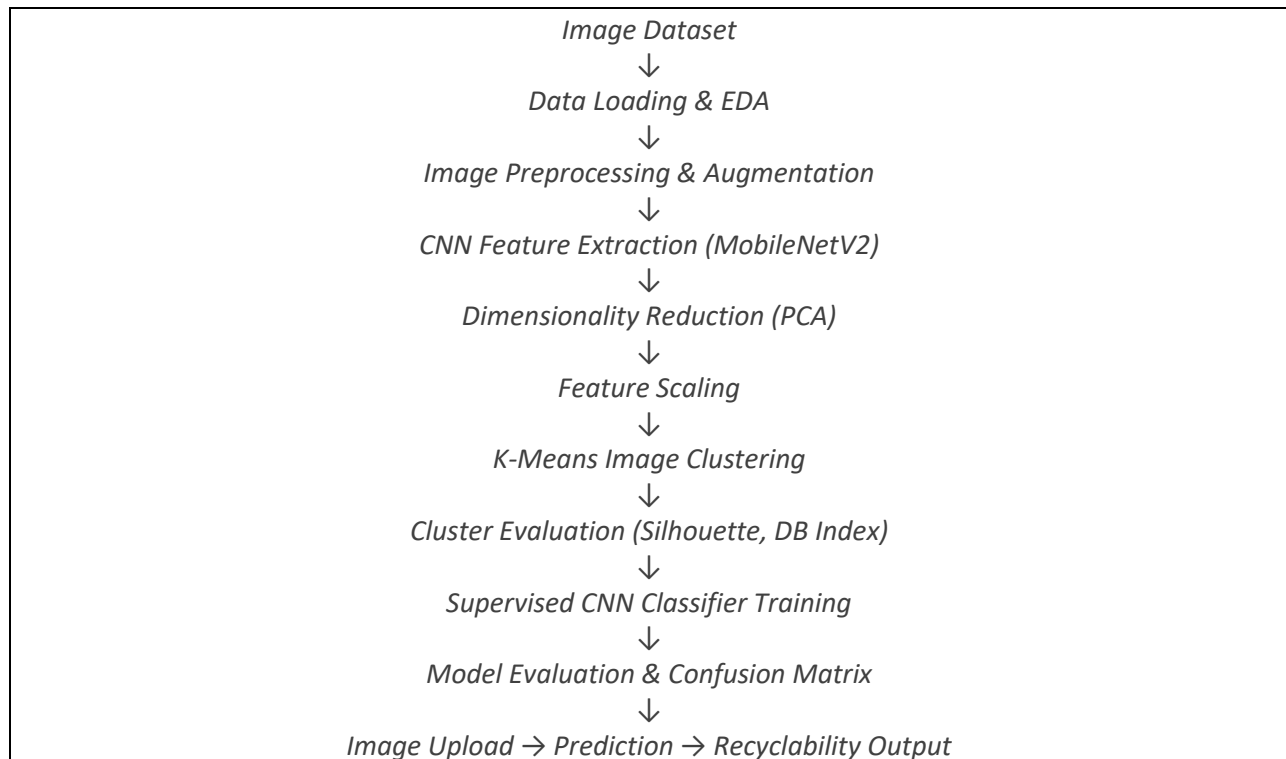
In this project we ai to create a waste classification system that classifies waste into different categories and predicts if it is recyclable or not using deep learning techniques and clustering.

The project combines image clustering using K-means, Feature extraction using CNN, dimensionality reduction using PCA and supervised classification to achieve reliable results.

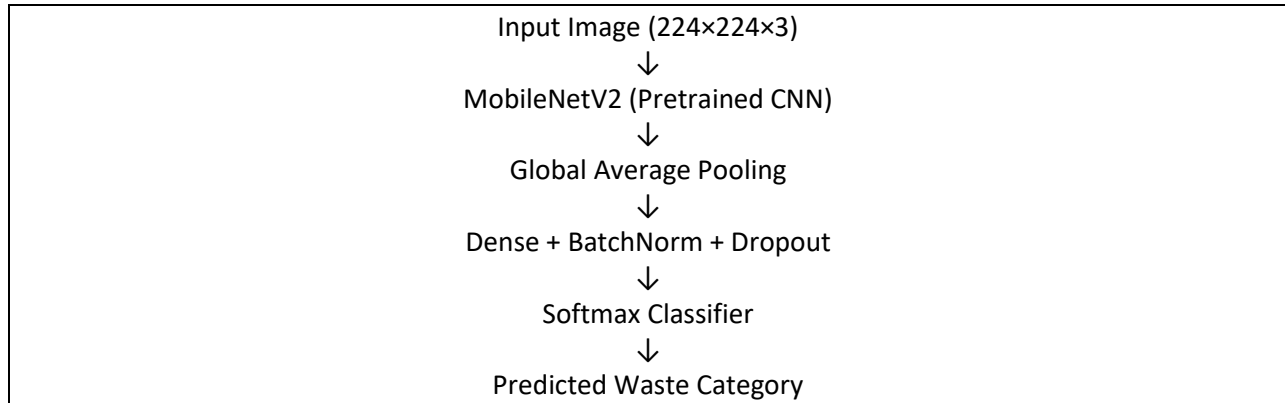
2 SYSTEM ARCHITECTURE

2.1 PIPELINE

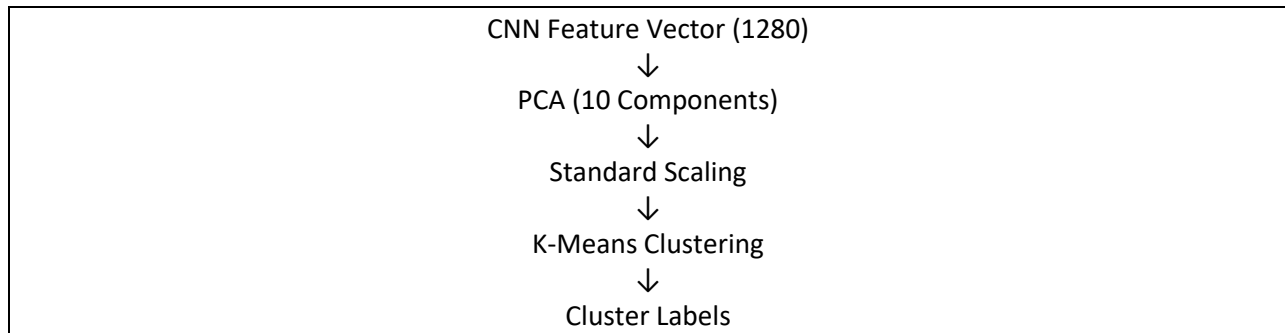
The complete pipeline of the project consists of the following stages:



ARCHITECTURAL DIAGRAM:



FOR CLUSTERING:



3 CONCEPTS USED:

The table Below highlights what concepts have been used while creating this project:

S. No	Concept Used	How It Was Used
1.	Image preprocessing	Resizing, normalization, augmentation
2.	Transfer learning	MobileNetV2 pretrained on ImageNet
3.	Feature extraction	CNN embeddings for clustering
4.	Dimensionality reduction	PCA to reduce 1280 → 10 features
5.	Unsupervised learning	K-Means clustering
6.	Cluster evaluation	Silhouette Score, Davies–Bouldin Index
7.	Supervised learning	CNN-based image classification
8.	Model evaluation	Accuracy, loss curves, confusion matrix
9.	Model persistence	Saving/loading trained models
10.	Practical deployment	Image upload and prediction in Colab

4 DATASET DESCRIPTION

4.1 DATASET SOURCE:

The Dataset contains labeled images of waste sourced from Kaggle. Each folder represents a category of waste.

4.2 WASTE CATEGORIES:

the dataset contains 9 classes of waste. The number of images per class is also mentioned.

- 1-Cardboard 461
- 5-Miscellaneous Trash 495
- 7-Plastic 921
- 8-Textile Trash 318
- 9-Vegetation 436
- 2-Food Organics 411
- 4-Metal 790
- 6-Paper 500
- 3-Glass 420

4.3 EXPLORATORY DATA ANALYSIS:

EDA was conducted to:

- Identify number of classes.
- Number of images per class
- Data was visualized two ways.
- Literal visualization to check images present in the dataset.
- Graphical representation using a bar graph to show clear count of each class and to visualize class distribution making balance or imbalance clear between classes.

5 DATA PREPROCESSING:

5.1 IMAGE PREPROCESSING:

Following preprocessing steps were applied:

- Image resizing to 224×224
- Pixel normalization (rescaling by $1/255$)
- Convert color space from BGR to RGB

5.2 DATA AUGMENTATION:

To reduce overfitting and increase robustness, following augmentations were applied:

- rotations
 - Horizontal flips
 - Width and height shifts
 - Zoom augmentation
- the dataset was split into:
 - 80% training
 - 20%validation

6 FEATURE EXTRACTION USING CNN:

6.1 WHY?

Raw images are too large and complex for clustering, therefore a pretrained CNN is used to extract compact, meaningful features.

6.2 MODEL USED:

- MobileNetV2
- Pretrained on ImageNet
- Classification head removed
- Global Average Pooling applied

This produces a 1280-dimensional feature vector per image.

6.3 FINE-TUNING:

- Most CNN layers were frozen.
- Last 40 layers were unfrozen to adapt features to waste images.

7 DIMENSIONALITY REDUCTION USING PCA

7.1 WHY WAS PCA USED?

Clustering becomes difficult because CNN features are high dimensional and noisy due to curse of dimensionality.

PCA was applied to:

1. reduce noise
2. speed up clustering
3. improve clustering metrics
4. improve cluster separability

5. to reduce DB score to a good range
6. to increase silhouette to a good range

7.2 PCA CONFIGURATION

- Original features: 1280
- Reduced features: 10
- Selected based on best clustering performance
- Selected among three more candidates, 20, 30 and 40 features

8 FEATURE SCALING

StandardScaler was applied to:

1. Normalize feature distributions
2. Ensure Equal contributions of all features
3. Improve K-means Distance calculations

9 IMAGE CLUSTERING USING K-MEANS

9.1 ALGORITHM USED

- K-Means Clustering
- Number of clusters = Number of waste classes which is 9.
- Initialization count increased to 20 for stability

9.2 CLUSTER EVALUATION METRICS

Two evaluation metrics used

- Silhouette Score = 0.14797406 (higher is better)
- Davies-Bouldin Score = 1.7074864541619457

The best version of these matrices was achieved using PCA with 10 components.

10 IMAGE CLASSIFICATION MODEL

10.1 MODEL DESIGN

The classification model consists of:

- MobileNetV2 base (frozen)
- Global Average Pooling
- Batch Normalization

- Dense layer (256 neurons)
- Dropout (0.5)
- Softmax output layer

10.2 HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Batch Size	32
Epochs	20
Loss	Categorical Crossentropy

10.3 TRAINING STRATEGY

- Early stopping so that the model does not overfit.
- Validation accuracy used to save the best model.

11 EVALUATION RESULTS

11.1 ACCURACY AND LOSS

Train and validation graphs show:

- Stable convergence
- Limited overfitting
- Reasonable generalization

11.2 CONFUSION MATRIX

- Strong performance for classes that are visually less similar such as glass and vegetation.
- Confusion between similar looking classes such as paper vs cardboard, Plastic vs Miscellaneous due to packaging, color or texture.

11.3 CLASSIFICATION REPORT

Overall accuracy can be estimated to 65% approximately, which is reasonable because:

- Class Imbalance is present in the dataset.
- Real world images vary greatly.
- There are visual similarities between waste classes.

12 IMAGE UPLOAD AND PREDICTION

The system allows us to:

- Uploading an image via Google Collab
- Preprocessing the image
- Predicting its waste class
- Determining if it is recyclable or not, or if it's biodegradable.
- Displays the image with the prediction made.

13 CHALLENGES AND LIMITATIONS

CHALLENGES:

- waste categories were visually similar thus separable clusters became difficult to achieve.
- CNN features were high dimensional which was not suitable for images of waste.
- Classes of waste were imbalanced.
- The clustering boundaries were weak when working with raw features.

LIMITATIONS:

- Model not optimized for real-time mobile deployment.
- Model accuracy depends on the quality of the dataset.
- K-means assumes the clusters to be spherical
- PCA may remove some fine details.

14 FAIRNESS AND ETHICAL CONSIDERATIONS

- Deployment should include confidence intervals
- Incorrect predictions will affect recycling decisions
- It is possible that the dataset may be biased towards one type of waste
- Model should assist humans and it should not replace decision making

15 SETUP AND RUN INSTRUCTIONS

15.1 ENVIRONMENTAL SETUP

Google Collaboratory is recommended as the platform

Required Libraries:

- Tensorflow
- Scikit-learn
- Matplotlib

- Seaborn
- Opencv
- Python

15.2 HARDWARE SETUP

- Minimum 8 GB RAM
- GPU Recommended (Colab GPU will be sufficient)

15.3 RUNNING THE PROJECT

1. Mount Google Drive
2. Unzip the dataset in Colab
3. Run the notebook cells in sequence
4. Train the model
5. Upload image for prediction

16 CONCLUSION

This project shows how Deep learning and clustering can be used to build a practical and working waste management project. This project is of low efficiency and a simpler version to a very complex requirement yet it gives a clear view into what could be made possible with more improvements. The project integrates both supervised and unsupervised learning and achieves meaningful clustering in the presence of overlapping real world images.